

How to install java & set path & check version

1. Installing Java

Step 1: Download Java Development Kit (JDK)

1. Visit the [Oracle JDK download page](#) or the [OpenJDK download page](#).
 2. Choose the version you need (typically JDK 11 or JDK 17, as these are widely used and LTS versions).
 - **1Z0-811 uses Java SE 11 as the target version.**
 3. Download the appropriate installer for your operating system (Windows, macOS, or Linux).
-

Step 2: Install the JDK

For Windows

1. Run the downloaded `.exe` file.
2. Follow the installation wizard:
 - Accept the license agreement.
 - Select the installation directory (default is `C:\Program Files\Java\jdk-version`).
3. Finish the installation.

For macOS

1. Open the downloaded `.dmg` file.
2. Drag the JDK icon to the Applications folder.
3. Verify the installation using the Terminal.

For Linux

1. Use the terminal to install OpenJDK:

```
sudo apt update
sudo apt install openjdk-11-jdk
```

Replace `11` with the desired version if needed.

2. Verify the installation with:

```
java -version
```

2. Setting the PATH Environment Variable

Why Set the PATH?

The PATH environment variable allows you to run Java commands (`java` , `javac` , etc.) from any directory in the terminal or command prompt.

Step 1: Locate the JDK Installation Path

- Default paths:
 - **Windows:** `C:\Program Files\Java\jdk-version\bin`
 - **macOS/Linux:** `/Library/Java/JavaVirtualMachines/jdk-version/Contents/Home/bin`
-

Step 2: Add JDK to PATH

For Windows

1. Open **System Properties**:
 - Press `Win + Pause` > Click **Advanced system settings** > Click **Environment Variables**.
2. Under **System Variables**, find `Path` and click **Edit**.
3. Add the JDK `bin` directory path (e.g., `C:\Program Files\Java\jdk-version\bin`).
4. Click **OK** to save and close all dialogs.

For macOS/Linux

1. Open the terminal and edit the shell configuration file:
 - For **bash**: `~/.bash_profile` or `~/.bashrc`
 - For **zsh**: `~/.zshrc`
2. Add the following line:

```
export PATH=$PATH:/path/to/jdk/bin
```

Example:

```
export PATH=$PATH:/Library/Java/JavaVirtualMachines/jdk-11/Contents/Home/bin
```

3. Save the file and reload it:

```
source ~/.bash_profile
```

3. Verifying the Installation

Check Java Version

1. Open the terminal (macOS/Linux) or command prompt (Windows).
2. Run the following command:

```
java -version
```

- You should see the installed Java version, for example:

```
java version "11.0.18" 2024-01-16 LTS
Java(TM) SE Runtime Environment 18.9 (build 11.0.18+10-LTS)
Java HotSpot(TM) 64-Bit Server VM 18.9 (build 11.0.18+10-LTS, mixed mode)
```

Check Compiler (javac) Version

1. Run the following command:

```
javac -version
```

- Output example:

```
javac 11.0.18
```

4. Testing the Installation with a Simple Program

Step 1: Create a Java Program

1. Open a text editor (e.g., Notepad, VS Code).
2. Write the following program and save it as `HelloWorld.java`:

```
public class HelloWorld {  
    public static void main(String[] args) {  
        System.out.println("Hello, Java is installed successfully!");  
    }  
}
```

Step 2: Compile the Program

1. Open the terminal or command prompt.
2. Navigate to the directory where `HelloWorld.java` is saved.
 - Example:

```
cd C:\Users\YourName\Documents
```

3. Compile the program:

```
javac HelloWorld.java
```

- This creates a `HelloWorld.class` file.

Step 3: Run the Program

1. Execute the compiled program:

```
java HelloWorld
```

2. Output:

Hello, Java is installed successfully!

Troubleshooting Common Issues

Issue 1: `java` or `javac` Not Recognized

- Ensure the PATH variable is set correctly.
- Check if you installed the correct version of Java (x86 vs. x64).

Issue 2: Incorrect Java Version

- Use the `where java` command (Windows) or `which java` (macOS/Linux) to check the Java executable path.
- Update the PATH variable to point to the desired JDK version.